

FAST– National University of Computer and Emerging Sciences

DS2001 Introduction to Data Science

Fall 2025

Assignment#2 (Deadline: 2nd November 2025)

- ✓ **Activity Type: Group Project + Presentation**
- ✓ **Group Size: 3 students per group**
- ✓ **Duration: 10 days (Presentation schedule will be uploaded after the deadline)**
- ✓ **Presentation Time: 20 minutes per group**

Assignment Tasks

You will work in groups of 3 to:

Select one topic from the suggested topics. Collect or use an existing dataset. Implement the algorithm in Python. Explain the **mathematical/statistical concepts** that drive the algorithm. Analyze and visualize your results. Present your findings to the class.

Suggested Topics / Algorithms

- ◆ Gaussian and Bernoulli Naive Bayes
- ◆ Decision Tree and Random Forest Regression
- ◆ MultiNomial Logistic Regression
- ◆ Support Vector Machines

Presentation

- ◆ Explain what *your topic* means. Describe where your chosen algorithm fits in. Highlight 2–3 real-world examples of its use. You must explain the key math/statistics behind your algorithm. Use Python to demonstrate step-by-step results (Input data, Algorithm output, Visualizations (2D/3D plots)). What patterns or structures did you discover? Suggest potential real-world applications. Reflect briefly on your group's learning process.

Deliverables:

- ◆ Visual aids (slides, charts, animations) are a bonus.
- ◆ Code notebook (.ipynb) and Presentation slides (.pptx or .pdf) covering: Dataset description, Algorithm steps, Key mathematical ideas, Insights/conclusions

Evaluation Criteria

- ◆ After all presentations, each group will rank the other groups (excluding their own) on a **1–5 scale** in each category described below: (I will moderate and adjust for fairness if needed).

Evaluation Aspect	Description	Weight
Clarity & Structure	Presentation flow, clear explanations	20%
Mathematical Understanding	Correctness and clarity of formulas/concepts	25%
Practical Implementation	Quality of code, visuals, and analysis	25%
Creativity & Insight	Interpretation, or presentation style	20%
Engagement & Communication	Confidence, Q&A handling, group participation	10%